

Role of the clotting system in the pathogenesis of neuroimmunologic disease^{1, 2}

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Experimental allergic encephalomyelitis (EAE) is a prototypic autoimmune inflammatory disease of the central nervous system (CNS) (2-4, 27, 29). The disease is induced at will in rats and other laboratory animals by sensitization to CNS tissue, isolated white matter myelin, or myelin basic protein (MBP) combined with an appropriate immunologic adjuvant such as complete Freund's adjuvant (CFA) (2, 3, 27, 29). EAE is characterized clinically by paralytic signs and histopathologically by focal perivascular cellular infiltrates accompanied by varying degrees of myelin injury (2, 3, 27, 29). Because EAE may exhibit a remitting-exacerbating type of clinical course with development of plaque-like areas of demyelination, the experimental disease has attracted worldwide attention as a promising model system for better understanding of chronic demyelinating disorders of humans, as exemplified by multiple sclerosis (MS) (23, 24, 29, 30).

Considerable evidence exists that activated T lymphocytes, reactive with myelin in general and MBP in particular, represent an important immune effector mechanism responsible for the inflammatory events underlying EAE and MS (2-4, 45). There also is evidence that B cells and antibody have a role in the pathogenesis of both neurologic disorders (11, 15, 37, 40, 47, 48). Two closely related and as yet unresolved issues concerning EAE and MS are: 1) the anatomic site(s) at which the host immune effector (or effectors) initially affects brain and spinal cord target tissue and 2) the specific events following this initial effector-target in-

teraction that result in the development of neurologic disease.

The cerebrovascular interface separating the CNS compartment from immune effector moieties in the systemic circulation seems to be the most likely locus for initial effector-target interaction. Indeed, many studies have demonstrated that an abrupt increase in permeability of the normally restrictive cerebrovasculature, constituting part of the blood-brain barrier (BBB), is one of the early events in the development of EAE (see ref 16 for a recent example and ref 21 for an overview of published studies). Alterations in the BBB also have been reported to accompany acute clinical ex-

acerbations of MS (1, 23). At least two lines of evidence implicate cerebrovascular endothelial cells as the probable loci for binding of MBP-reactive effector cells. First, immunoreactive

ABSTRACT

Experimental allergic encephalomyelitis (EAE) is a prototypic neuroautoimmune disease involving sensitization to central nervous system myelin basic protein (MBP). Our studies of the clotting system and ensuing fibrinolysis implicate coagulation and cleavage of fibrin within or on the luminal surface of the cerebrovasculature as events initiating the inflammation characterizing EAE. Among recipient rats injected with MBP-primed, cultured-activated lymph node cells, opening of the blood-brain barrier (BBB) and deposition of perivascular fibrin within the spinal cord occur in parallel 1 day before onset of clinical signs of EAE. Daily treatment of recipient rats with *trans*-4-(aminomethyl)cyclohexanecarboxylic acid, a synthetic product that specifically inhibits plasminogen activator derived from endothelial cells, results in marked reduction of increased permeability of the BBB and suppression of clinical signs of EAE. We postulate that the critical event precipitating EAE is binding of circulating MBP-reactive immune effector cells to MBP immunodeterminants on the surface of cerebrovascular endothelial cells. Coagulation and ensuing fibrinolysis occur at sites of binding of effector cells to cerebrovascular endothelium. Release of biologically active peptides cleaved from fibrin open the BBB, thereby setting the stage for the cascade of inflammatory events culminating in clinical manifestations of EAE.—Paterson, P. Y.; Koh, C.-S.; Kwaan, H. C. Role of the clotting system in the pathogenesis of neuroimmunologic disease. *Federation Proc.* 46: 91-96; 1987.

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TABLE 1. BBB permeability and CNS fibrin deposits in normal rats and recipient rats injected with MBP-primed, activated donor LNC

Status of rats	Spinal cord segment	BBB permeability ^a	Fibrin deposits ^b
Normal	Cervical	1.06 ± 0.65	0
	Thoracic (lower)	1.03 ± 0.72	0
	Lumbosacroccygeal	1.14 ± 0.54	0
LNC recipient ^c	Cervical	5.15 ± 2.11	65 ± 37
	Thoracic (lower)	9.56 ± 6.81	108 ± 70
	Lumbosacroccygeal	16.04 ± 5.07	192 ± 74

^a BBB permeability was assessed with ¹²⁵I- and ¹³¹I-labeled albumin by the method of Leibowitz and Kennedy (20), with rats killed on the day of onset of EAE signs, and counts per minute per cord segment were expressed as extravascular blood equivalents (EVBE); mean EVBE values (±1 SD) were recorded for each segment of cord. ^b Recipient rats were killed 1 or 2 days after onset of clinical signs of EAE; mean number (±1 SD) of discrete fibrin deposits recorded for each segment of cord. ^c Recipient rats were injected i.v. with 3–10 × 10⁷ MBP-primed, cultured-activated LNC.

MBP, detected by sensitive radioimmunoassays and probably representing catabolic cleavage fragments, has been shown to partition between the cerebrospinal fluid and the systemic circulation in normal rats and clinically well humans (8, 10, 33, 34–36). Indeed, it has been our contention that CNS-derived MBP degradation fragments that have emigrated into the systemic circulation may serve as neuroimmunoregulators. These fragments restrain MBP-reactive cells, which exist in clinically well animals and humans (6, 14, 49), from undergoing clonal activation, which leads to neuroimmunologic disease (29, 33). Second, MBP is demonstrable within cerebrovascular endothelial cells of mice that develop EAE (43).

In both EAE and MS, increased BBB permeability is temporally associated with accumulations of fibrin within and around CNS blood vessels (1, 28, 31, 36). Most investigators have viewed these perivascular deposits of fibrin as little more than a secondary patent reflection of increased BBB permeability without etiopathogenetic significance in terms of the concomitant evolving CNS inflammatory reaction. In contrast, our observations have increasingly implicated activation of the clotting system, as manifested by perivascular fibrin deposition, as an important early event in expression of clinical neurologic signs of EAE (28, 31, 38). Evidence leading to this thinking was the basis for our discussions in both of the preceding symposia of this series (28, 31).

This paper extends the theme stated above by summarizing additional data secured through work with the cell-transferred form of EAE in

isohistogenic Lewis rats. Our observations lead to two postulates: 1) EAE effector cells directly cause increased BBB permeability through activation of the clotting cascade and resulting fibrinolysis, and 2) opening of the BBB is the prerequisite event setting the stage for EAE and all inflammatory events characterizing this neuroimmunologic disease.

METHODOLOGICAL DETAILS

A big advantage of successful transfer of EAE by means of sensitized donor cells (26) is that late immune effector responses involving CNS target tissues can be isolated for study. We used the activated cell transfer system described by Panitch and McFarlin (25) and Reichert et al. (39), in which Lewis rat lymphoid cells primed with MBP-CFA are cultured with MBP for 3 days. During this time they acquired a remarkably increased capacity to transfer EAE to normal syngeneic Lewis recipient rats. In all the experiments summarized in this paper, normal Lewis rats were injected i.v. with 3–4 × 10⁷ lymph node cells (LNCs) that had been collected 9 days after priming donors with MBP-CFA and cultured for 72 h with MBP as described by Mannie et al. (23). In each experiment, virtually all recipients in each control group of animals (usually there were at least four rats per group) developed clear-cut neurologic signs ranging from atonia of the tail to complete paralysis of both hind legs. In every instance where histological studies were carried out, characteristic inflammatory lesions of EAE were readily demonstrable in the brain and spinal cord. Fibrin deposits were de-

tected in cryostat sections of spinal cord by direct immunofluorescence and quantitated within each segment of cord as described previously (28, 31, 36). Alterations in BBB permeability were quantitated by a double-isotope method (20). Partitioning of ¹²⁵I- and ¹³¹I-labeled albumin, injected i.v. 20 h and 5 min, respectively, before the rats were killed, reflected BBB permeability and was expressed in terms of extravascular blood equivalents (EVBE) (20). Focal areas of fibrinolysis within the lumbosacroccygeal (LSC) segment of the spinal cord were detected and quantitated by the fibrin slide technique (17, 18).

BBB PERMEABILITY CHANGES IN TERMS OF ONSET OF EAE

Several experiments are summarized in Table 1. Among normal rats, very little ¹³¹I- or ¹²⁵I-labeled albumin traversed the cerebrovasculature and fibrin deposition was not demonstrable within the spinal cord. In contrast, there were a readily detected increase in BBB permeability and conspicuous perivascular deposits of fibrin within the CNS of LNC recipients that developed EAE. Increased BBB permeability and deposition of perivascular fibrin occurred in parallel and were most marked in the LSC segment of spinal cord. Other studies along this line revealed that BBB alteration and fibrin deposition were first detectable 1 day before onset of clinical neurologic signs, i.e., on day -1. Maximum values were attained on day 0, which was designated as the day of onset of signs in individual rats (36).

EFFECT OF A FIBRINOLYTIC INHIBITOR ON BBB PERMEABILITY AND EAE

We next turned to the role of the clotting system and fibrinolysis in the context of the observations summarized in Table 1 and our previous work on the relationship of the coagulation system and EAE (28, 31, 36). Brosnan et al. (5) and Smith and Amaducci (41) reported that several drugs capable of inhibiting fibrinolysis had a suppressive effect on development of EAE in Lewis rats actively sensitized to MBP-CFA. We assessed one of these drugs, the synthetic fibrinolytic inhibitor *trans*-4-(aminomethyl)cyclohexanecarboxylic acid (AMCA), in terms of

TABLE 2. BBB permeability in AMCA-treated and control recipient rats injected with MBP-primed, activated donor LNC

Status of recipients	BBB permeability ^a	
	Lower thoracic	LSC
Saline-treated	13.6 ± 7.2	18.5 ± 3.5
AMCA-treated ^b	3.2 ± 1.0 ^c	8.5 ± 3.3 ^d

^a BBB permeability was assessed as described in footnote a of Table 1. ^b AMCA treatment began 2 days after i.v. injection of 3–10 × 10⁷ MBP-primed, cultured-activated LNC, with i.p. injection of 50 mg AMCA every 12 h, and continued until control, saline-treated recipient rats developed clinical signs of EAE. ^c *P* < 0.05 compared to saline-treated recipients by a two-sample *t* test (two-sided). ^d *P* < 0.01 compared to saline-treated recipients by a two-sample *t* test (two-sided).

its capacity to interfere with BBB permeability changes in recipients injected with primed, activated donor lymphoid cells. Table 2 summarizes the results of several representative experiments. AMCA-treated recipient rats tended to have a much smaller increase in BBB permeability than control recipients. Seven separate experiments indicated that treatment with AMCA, under the experimental conditions described (see Table 2 and footnotes), exerts a statistically significant effect with respect to quenching or diminishing BBB permeability. Diminished BBB permeability coincided with reduction in or abrogation of clinical neurologic signs (36).

Of considerable interest to us was the abrupt and remarkable decrease in cerebrovascular endothelial cell-associated fibrinolysis among non-drug-treated cell recipients in relation to onset of clinical signs of EAE. Pertinent data, as set out in Table 3, indicate that within 1 day before the onset

TABLE 3. Fibrinolysis in LSC spinal cord of normal rats and recipient rats developing EAE after injection of MBP-primed, activated donor LNC

Status of rats	Days before or after onset of EAE signs	Mean fibrinolytic score ^a
LNC recipient ^b	-1	43 ± 8
	0	11 ± 6
	+5	59 ± 9
Normal		107 ± 14

^a Mean number (±1 SD) of discrete fibrinolytic foci in two separate sections of LSC cord of each recipient rat in each group of LNC recipients and of several normal rats. ^b Each recipient rat in each group of animals was injected with 40 × 10⁶ MBP-primed, cultured-activated LNC.

of clinical signs, i.e., day -1, the number of fibrinolytic foci within the LSC segment of spinal cord of recipients is markedly smaller than the number of foci in normal rat cords. The nadir in fibrinolytic foci appears to occur on the day when clinical signs first appear. Thereafter, there is a slow increase in fibrinolytic activity. But, as shown in Table 3, even at 5 days after the onset of signs, a time when most recipient rats exhibit clinical recovery, the degree of fibrinolytic activity in no way approaches that of control, normal rats over the observational time frame depicted. Whether the sudden reduction of fibrinolytic foci, at least in part, represents depletion of cerebrovascular endothelial cell plasminogen activator (PA) content or reflects other mechanisms, such as inhibition of PA activity by mobilization of autologous PA inhibitors, is a question we are currently addressing.

SEQUENCE OF EVENTS IN NEUROIMMUNOLOGIC INFLAMMATORY DISEASE

In terms of EAE as a model disease, the most likely sequence of effector events leading to CNS inflammation and culminating in neuroimmunologic disease is listed in Table 4. First, there is production of high-affinity binding antibody reactive with circulating, endogenous MBP fragment(s). Production of such antibodies occurs in response to intense stimulation by injection of MBP emulsified in CFA (9, 27,

33). The circulating MBP-fragment acts as a neuroimmunoregulator that restrains MBP-reactive T cells from expressing their potential injurious anti-CNS effector immune activity (8, 33, 35, 36). As reviewed elsewhere (33), free anti-MBP high-affinity binding antibody has been observed in only three situations to date: 1) rats developing EAE, 2) patients exhibiting acute exacerbations of MS, and 3) a single patient with postinfectious acute disseminated encephalomyelitis. Preferential activation of B cells occurs in response to sensitization with MBP-CFA because they are generally less responsive or tolerant to self-antigens than are T cells, as shown by the studies of Weigle and associates (46). In MS patients, the stimulus for MBP-reactive B cell activation and clonal expansion may be persistent infection by a virus recognized as nonself by T helper lymphocytes; interaction of endogenous MBP with receptors on nontolerant MBP-reactive B cells together with help provided by the reacting T helper cells would set the stage for B cell activation and production of antibody specifically reactive with circulating MBP neuroimmunoregulator.

Second, rising titers of anti-MBP antibody and complexing of such antibody with circulating MBP fragment(s) would clear the neuroimmunoregulator from the circulation (44).

Third, removal of circulating neuroimmunoregulator would allow MBP-reactive T cells to become acti-

TABLE 4. Specific sequence of events comprising the inflammatory pathway of EAE in Lewis rats

- 1) Production of high-affinity binding antibody reactive with circulating MBP fragment(s) acting as neuroimmunoregulator
- 2) Clearing of high-affinity MBP neuroimmunoregulator from the circulation
- 3) Entry into the circulation of MBP-reactive EAE effector cells
- 4) Binding of effector cell to EAE determinant on CNS-derived MBP fragment on surface of cerebrovascular endothelial cell
- 5) Activation of effector cell with secretion of procoagulant, initiating clotting and deposition of fibrin on surface of endothelial cell
- 6) Activation of endothelial cell, release of PA, initiation of plasmin-mediated fibrinolysis with cleavage of peptides from fibrin mediating increased BBB permeability and chemotaxis
- 7) Fluid influx and emigration of inflammatory cells into the CNS compartment:
 - a) Macrophage-induced injury of myelin membrane resulting in perturbation of propagation of nerve impulses along axon fibers
 - b) Altered axonal conductance resulting in paralytic neurologic signs
- 8) Release of high-affinity neuroimmunoregulator MBP fragment(s) into the circulation with reestablishment of MBP tolerance
- 9) Depletion of endothelial cell PA and resulting cessation of fibrinolysis leading to closing of the BBB and termination of the EAE inflammatory cascade
- 10) Dissipation of intra-CNS inflammation and repair of myelin membrane injury, with return of axonal conductance, leading to clinical recovery

vated and to be demonstrated in peripheral lymphoid tissues (9). In the absence of circulating MBP fragment(s), activated MBP-reactive effector T cells would soon be demonstrable in the peripheral blood.

Fourth, circulating MBP-reactive effector T cells, no longer tolerant to MBP as a consequence of removal of endogenous MBP neuroimmunoregulator (see above), now could bind to the EAE determinant on CNS-derived MBP fragment(s) on the surface of cerebrovascular endothelial cells. Inasmuch as such cells possess class I major histocompatibility complex antigens, the prerequisites for interaction of effector T cells with CNS target cells would be satisfied. It should be noted that the effector cell postulated to interact with endothelial cells has not been identified. There is no reason a priori for excluding the possibility that the effector cell, in fact, is an anti-MBP antibody-producing B cell or a mononuclear cell expressing antibody-dependent cellular cytotoxicity.

Fifth, the interaction of effector cells with EAE determinant containing MBP-fragment(s) within or on the surface of endothelial cells leads to further activation of the effector cells and results in secretion of procoagulant by the activated cells. This action leads to initiation of clotting and deposition of fibrin on luminal surfaces of CNS target endothelial cells. Activation of the clotting system resulting from procoagulant secretion by macrophages, T cells, or B cells is a well-established parameter of effector cell-antigen binding (32). Geczy et al. (12) have shown that secretion of procoagulant by sensitized lymphoid cells is a direct correlate of cell-mediated immunity directed against MBP in animals susceptible to EAE. Our earlier work with rats rendered profoundly hypofibrinogenemic as a result of injections of anacrod indicated that activation of the clotting cascade is a prerequisite for development of clinical neurologic signs of EAE (28).

Sixth, microdeposits of fibrin on cerebrovascular endothelial cells would lead to activation of these cells and release of PA. In situ plasmin-mediated fibrinolysis, with cleavage from fibrin of biologically active peptides, namely permeability factor and chemotactic factor (32, 36), would set the stage for increased permeability of the BBB and emigration of inflammatory cells into the CNS compartment.

Seventh, sudden influx of fluid and migration of inflammatory cells, particularly macrophages, would be expected to result in physiological and biochemical perturbation of myelin lamellae enveloping axonal nerve fibers and to disturb nerve impulse propagation (42). Altered axonal conductance would translate to clinical neurologic abnormalities.

Eighth, sudden release of an excess of high-affinity MBP fragment(s) into the circulation occurs in rats in concert with the appearance of clinical signs of EAE (35). Entry of increased amounts of MBP fragment(s) would neutralize circulating high-affinity binding anti-MBP antibody and reestablish tolerance among MBP-reactive effector T cells.

Ninth, diminished fibrinolysis resulting from depletion of endothelial cell PA or other mechanisms would result in closing of the BBB and termination of the EAE inflammatory cascade.

Tenth, dissipation of the intra-CNS inflammation and resulting repair of myelin membrane injury would result in return of normal axonal nerve impulse conductance, with onset of clinical recovery (42).

In conclusion, the principal message of our observations is twofold. First, the initial and immunologically specific event, which initiates the inflammatory response and tissue injury that translate to EAE, involves binding of immune effector cells with an EAE-inducing immunodeterminant of MBP on the luminal side of the CNS vasculature. All events that occur there-

after are immunologically nonspecific, namely activation of clotting, fibrin deposition, fibrinolysis, opening of the BBB, and influx of inflammatory cells into the neuraxis with injury to the neurophil. Two observations support our premise that immune effector cells must interact with cerebrovasculature in some fashion for their EAE-causing potential to be expressed: 1) intracerebral injection of rats with sensitized donor LNC was not accompanied by clinical signs of EAE and did not result in shortening of the latent period before occurrence of typical lesions of the disease or alter the distribution of lesions in the neuraxis from that consistently observed after i.v. injection of donor cells (38), and 2) intrathecal injection of activated continuous line MBP-reactive T cells, in numbers that elicit paralytic signs of EAE when injected i.v., resulted in no clinical abnormalities (H. Lassmann, personal communication). To be sure, EAE-immune effector cells are present among the inflammatory cells comprising the lesions of EAE (7, 13). Further cycles of activation of such cells in situ may contribute to subsequent attacks of clinical disease characterizing the chronic, relapsing form of EAE (19). It is our contention, however, that immune effector cells need not penetrate the CNS compartment per se to initiate the first clinical episode of EAE. Second, opening of the BBB appears to be a prerequisite for the development of inflammatory neuroimmunologic disease, as exemplified by EAE. [P]

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Discussion

Dr. Swanborg: In one of your slides you showed a very dramatic change in the BBB in the lumbosacral region of the cord. Does this reflect what is known about the anatomy of the endothelium in that portion of the cord? Are there any differences known that would account for the changes being predominately in that region?

Dr. Paterson: These are important questions. There are at least two possibilities. First, lumbosacral cord endothelial cells may excel with respect to expressing Ia class II major histocompatibility complex gene products on their surface in response to interferon—or other Ia-inducing signals. If so, such cells might well be better endowed to present MBP antigen if, indeed, the presence of Ia is required for EAE-effector cell binding to cerebrovascular endothelium. Second, lumbosacral cord endothelial cells may possess an inherent capacity to place on their surface greater amounts of MBP catabolic fragment(s) containing encephalitogenic immunodeterminant, thereby causing circulating EAE effector cells to interact preferentially with these EAE determinant-bearing endothelial cells.

Dr. Jesmok: Do neutrophils play any role in the neurologic abnormalities produced?

Dr. Paterson: Under certain experimental conditions one can observe primarily a neutrophilic type of inflammatory reaction. In that case the neutrophil presumably causes the same kind of injury that macrophages elicit with respect to myelin. This injury occurs via a totally nonimmu-

nologically specific pathway, e.g., liberation of proteases.

Dr. Jesmok: Do the symptoms appear earlier in this case?

Dr. Paterson: I would like to rephrase my previous remark. There are different ways to influence the onset of EAE and the nature of the host inflammatory response. For example, you can add pertussis vaccine to the sensitization regimen and produce an accelerated and more severe form of EAE. Under these conditions there is an appreciable increase in the number of segmented neutrophils in the inflammatory response. As another example, the neonatal, suckling rat cannot mobilize macrophages effectively and its response is limited primarily to segmented neutrophils. Nevertheless, suckling rats 18, 20, or 22 days old develop a typical form of acute EAE without appreciable foreshortening of the latent period. The principal inflammatory cell in the tissue is the segmented neutrophil.

Dr. Vinegar: You could try to block the mobilization of each type of phagocytic cell specifically by injecting colchicine before their mobilization.

Dr. Paterson: By systemic treatment of the animal?

Dr. Vinegar: Yes. You may be able to stop the phagocytic cells from adhering to the endothelium by local injection of actinomycin D.

Dr. Paterson: Don't you think it might be easier to give one dose of dexamethasone that shuts the BBB on the one critical day?

Dr. Vinegar: Yes, but steroids have many pharmacological actions including at least two which are antiinflammatory. One of these actions is on the neutrophil in the circulation. This action reduces the ability of the neutrophil to adhere to the endothelium. Some of the other actions may complicate an interpretation of the results.

Dr. Ziff: You place the endothelial cell at the center of this response. You have it acting as an accessory cell for the antigen to sensitize the T cell. You suggest that it supplies the fibrinolysin, which digests the fibrin. Could it also supply the procoagulant that produces the fibrin at one stage in the process? It was demonstrated this week, I think by Dr. Stern at another session, that the endothelial cell does not produce procoagulant along with IL 1.

Dr. Paterson: Yes, papers in the literature suggest that endothelial cells can produce procoagulant. I did not say that the endothelial cell presents anything to T cells. That is a prevailing opinion. I think the endothelial cell is crucial in providing a specific determinant for the effector cell so that the effector cell knows where it is and to what specific target cell it must adhere to carry out its functions. I do not know whether it is necessary to have the antigen presented or for the antigen simply to be there. In the sequence of events I envision, the crucial event consists of effector cells "seeing" MBP-containing EAE-inducing immunodeterminant on surfaces of endothelial cells and binding to such cells irrespective of whether the cells express Ia or "present antigen."